

8B 实验结果分析报告

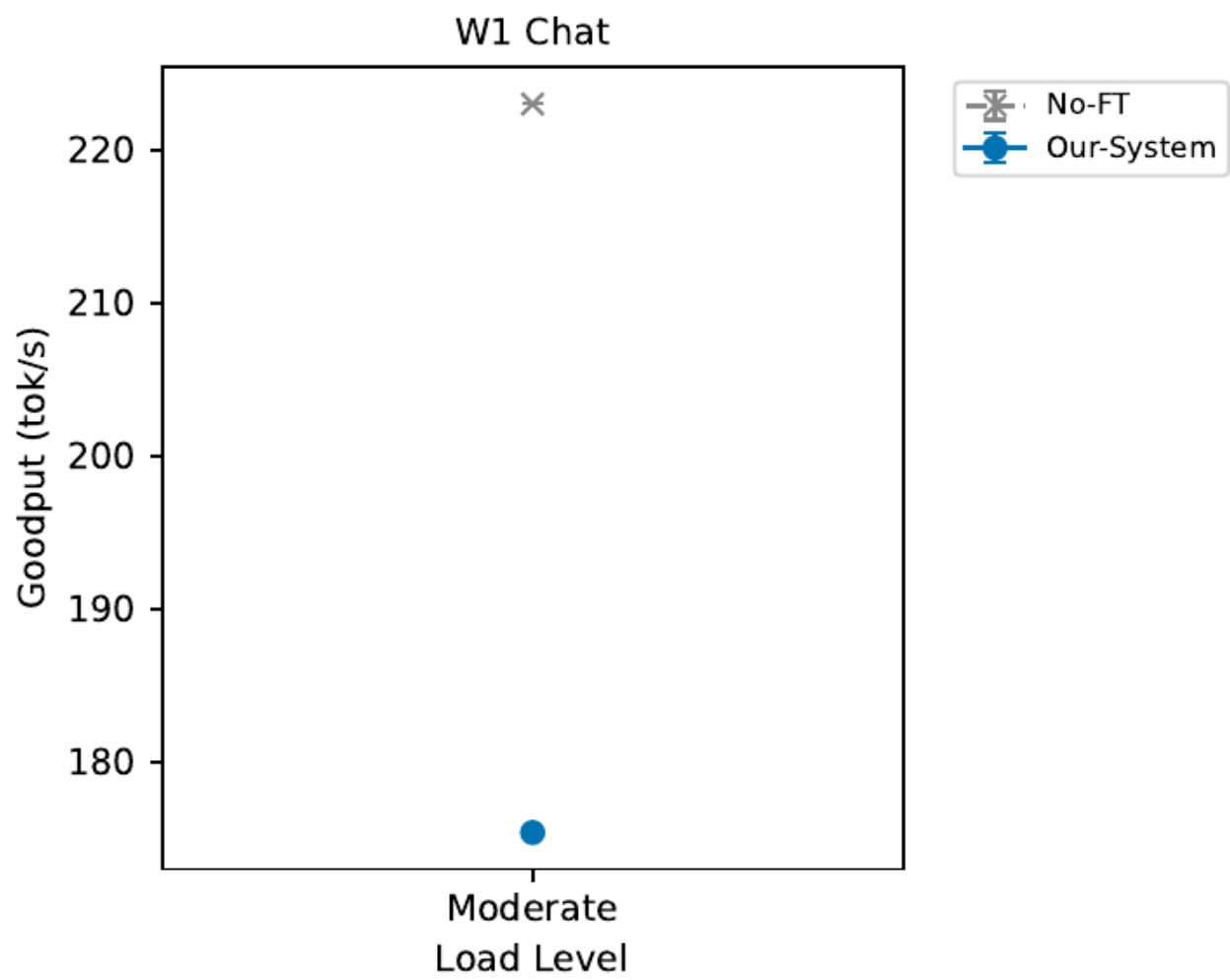
所有 bug 修复后的结果。Config: config_8b_calibrated.yaml, max_model_len=8192, dp=2, seed=42。

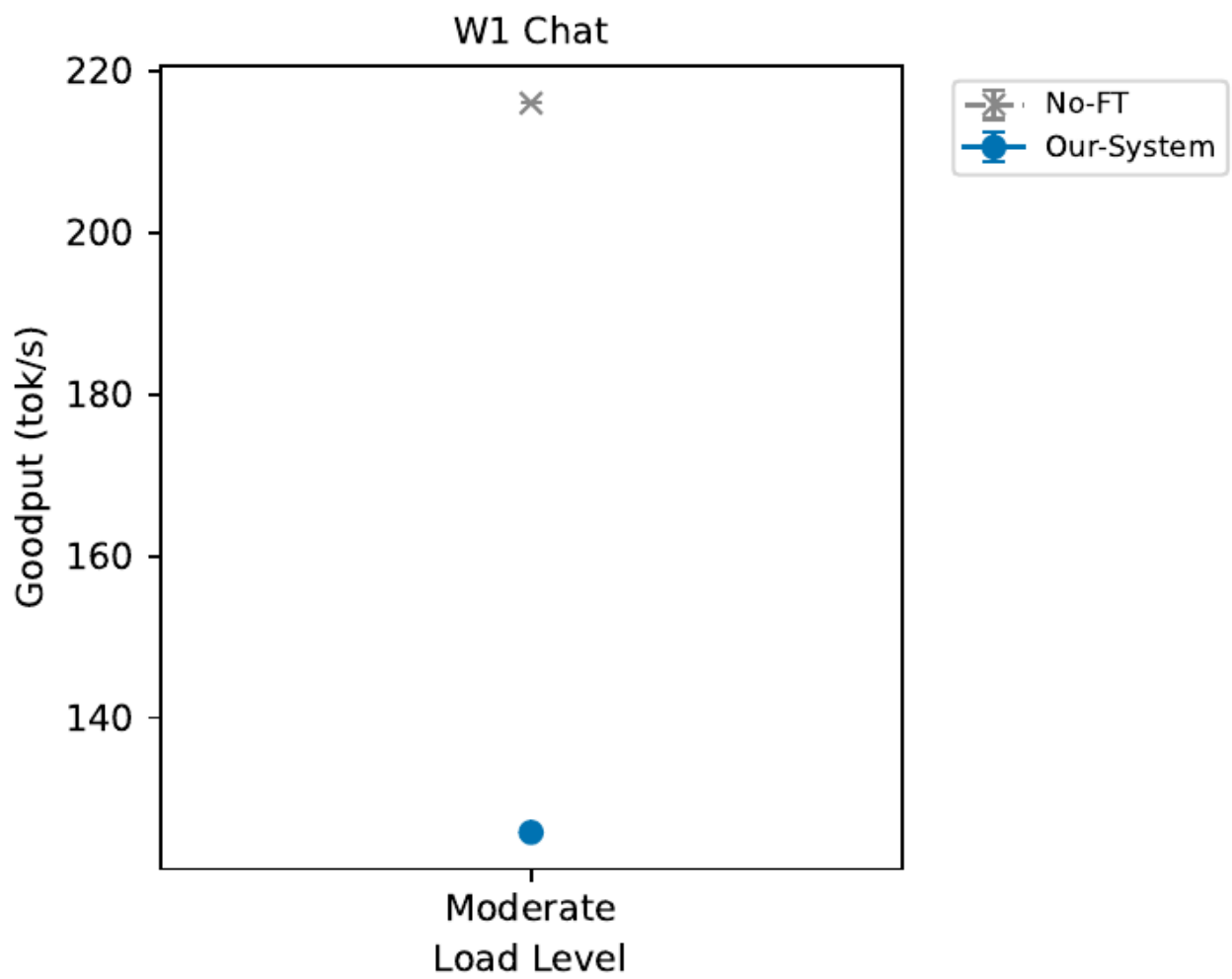
模型：Llama-3.1-8B-Instruct，硬件：2×NVIDIA RTX A6000 48GB。

1. E0_Smoke — Pipeline 验证 (4 runs)

Baseline	Fault	Goodput	完成率	SLO Viol	TPOT P95	恢复率
No-FT	none	223.1	98.8%	0.0%	29.9ms	—
No-FT	F2_Mid	216.0	96.8%	0.0%	37.8ms	0%
Our-System	none	174.8	98.4%	18.9%	80.3ms	—
Our-System	F2_Mid	120.5	98.4%	45.5%	125.2ms	100%

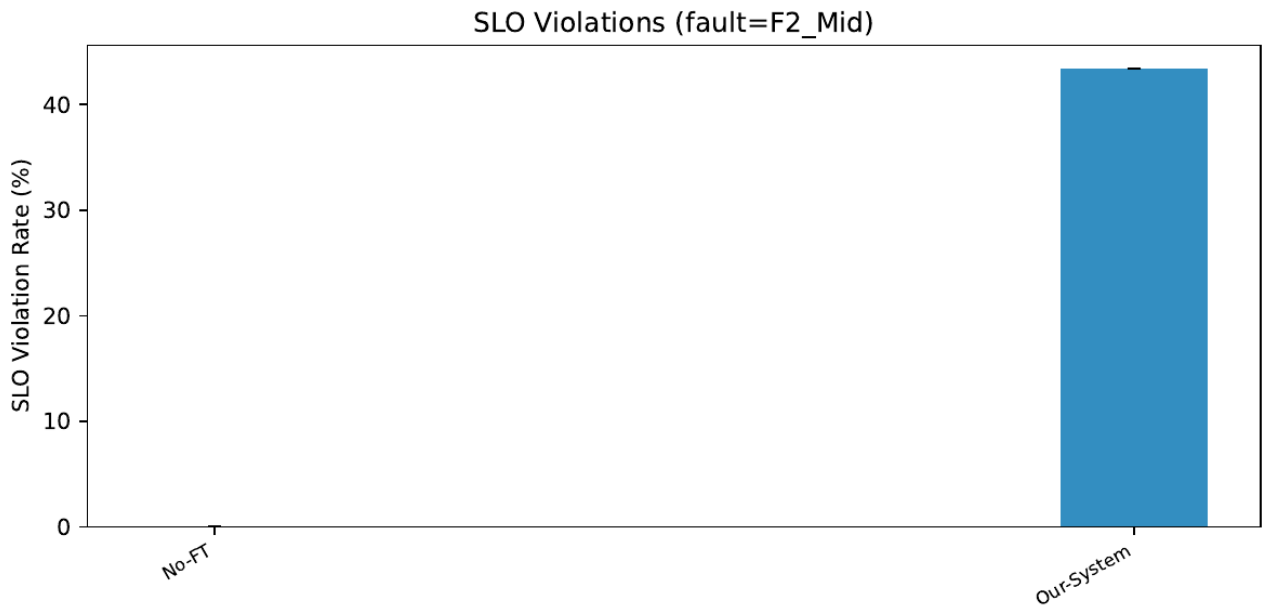
Goodput





SLO Violation





要点

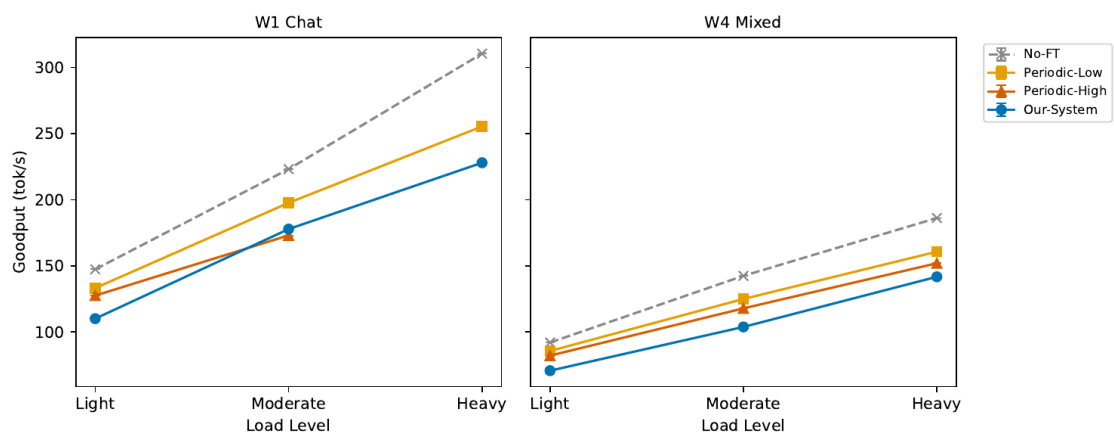
- Checkpoint 正常 publish, 恢复率 100%
- Our-System 正常态 goodput 比 No-FT 低 22% (FT 开销)
- SLO violation 高 (18-45%) —— TPOT 80-125ms 超过 50ms SLO

2. E1a_Quick — 端到端主结果 (48 runs, 1 failed)

47 ok, 1 failed (Periodic-High/W1_Chat/Heavy/none, prompt+output 超 max_model_len=4096, 已改为 8192)。

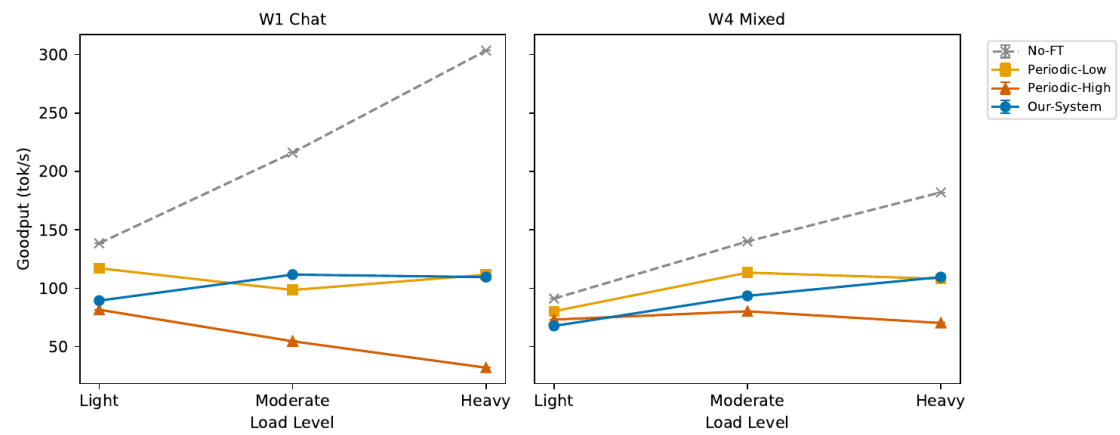
Goodput vs Load

无故障:



goodput_none

有故障 (F2_Mid):



goodput_f2mid

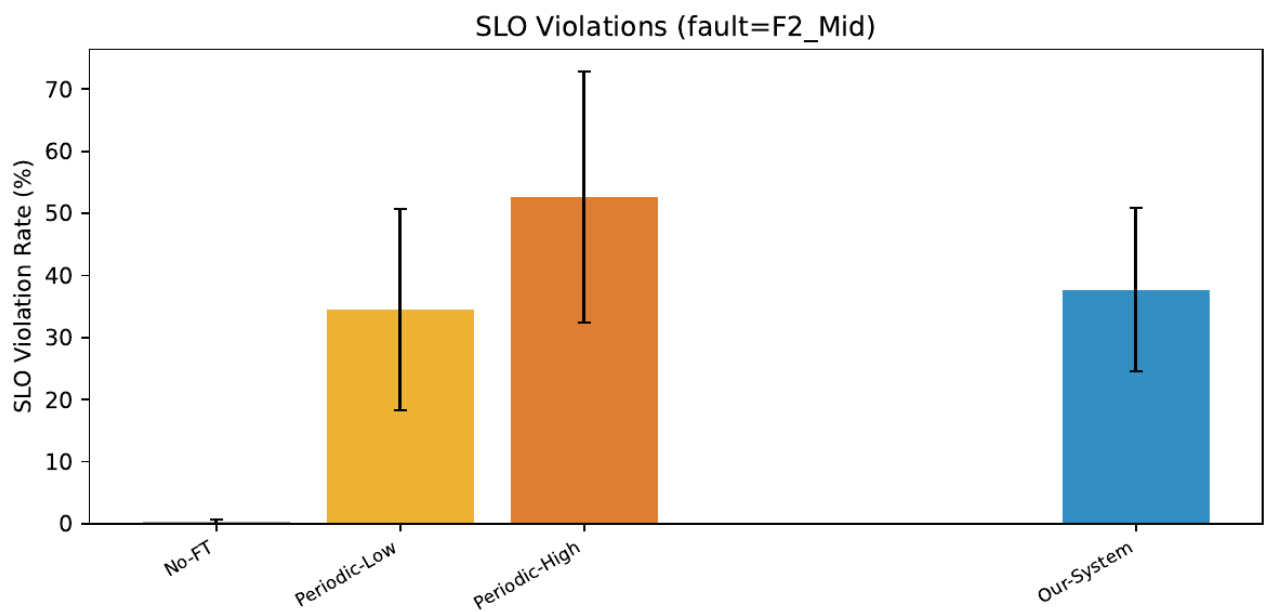
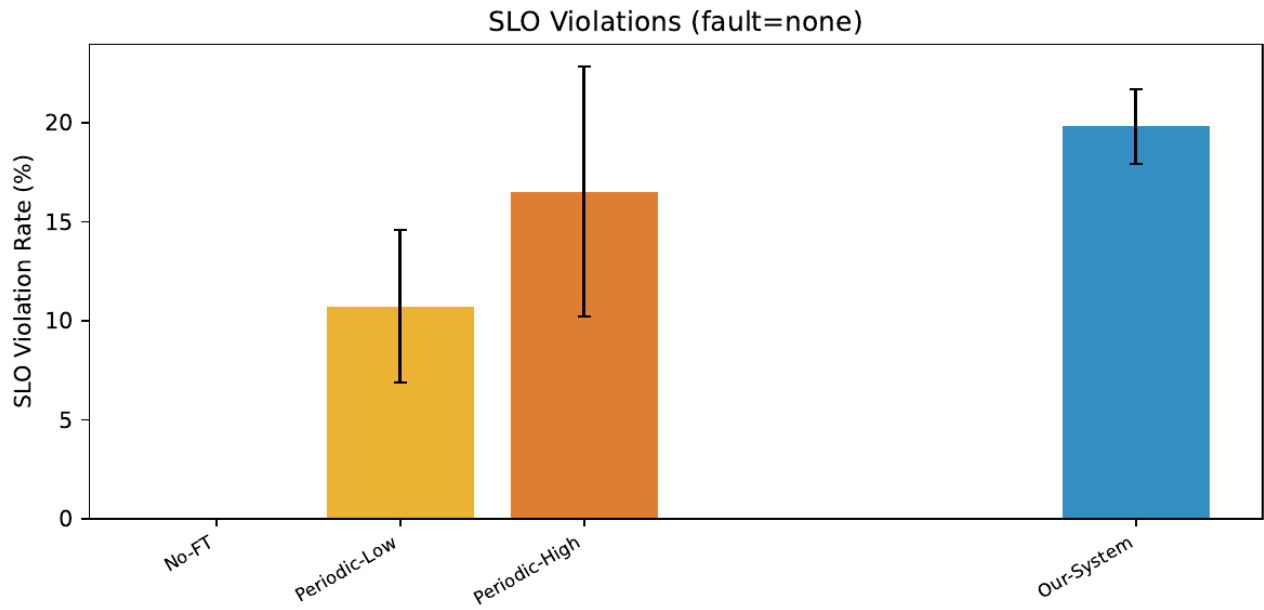
无故障 Goodput 数据

Baseline	W1_Chat Light	W1_Chat Mod	W1_Chat Heavy	W4_Mixed Light	W4_Mixed Mod	W4_Mixed Heavy
No-FT	147.3	223.1	310.4	92.0	142.4	186.1
Periodic-Low	133.2	197.6	255.3	85.6	124.9	160.7
Periodic-High	127.5	173.0	(fail)	82.1	117.8	151.9
Our-System	110.1	177.8	227.9	70.7	103.8	141.7

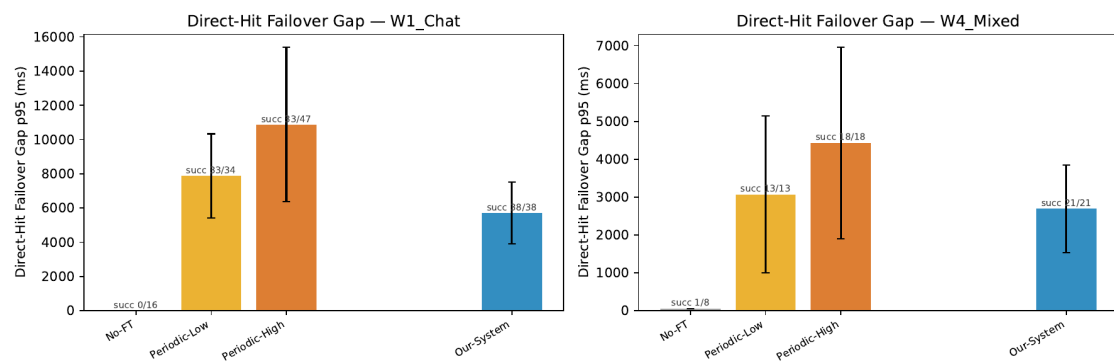
有故障 Goodput 数据

Baseline	W1_Chat Light	W1_Chat Mod	W1_Chat Heavy	W4_Mixed Light	W4_Mixed Mod	W4_Mixed Heavy
No-FT	138.5	216.0	303.4	91.1	140.1	182.1
Periodic-Low	117.1	98.5	111.5	80.3	113.4	108.0
Periodic-High	81.6	54.6	32.0	73.1	80.3	70.2
Our-System	89.4	111.7	109.5	67.7	93.5	109.5

SLO Violation

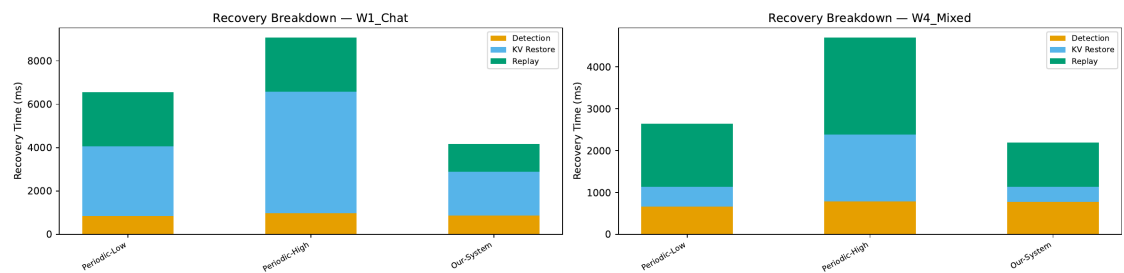


Failover Gap



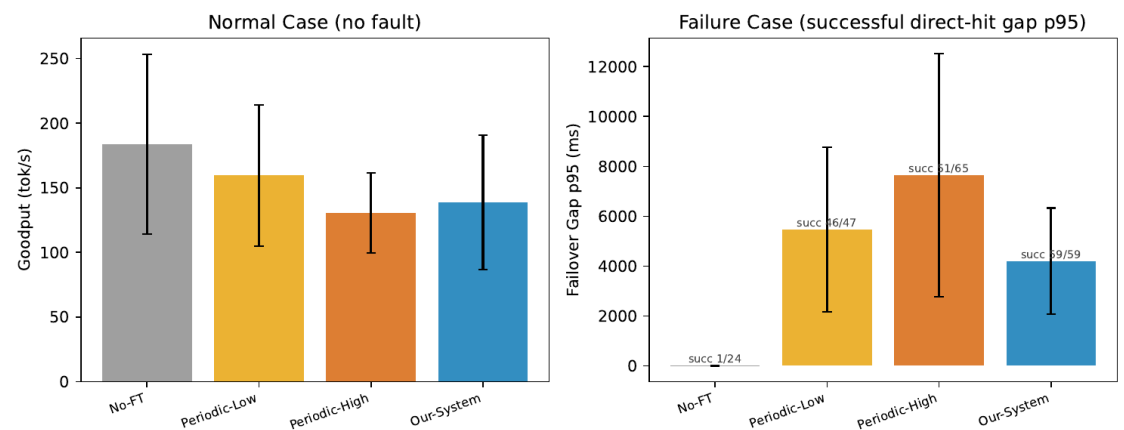
failover_gap

Recovery Breakdown



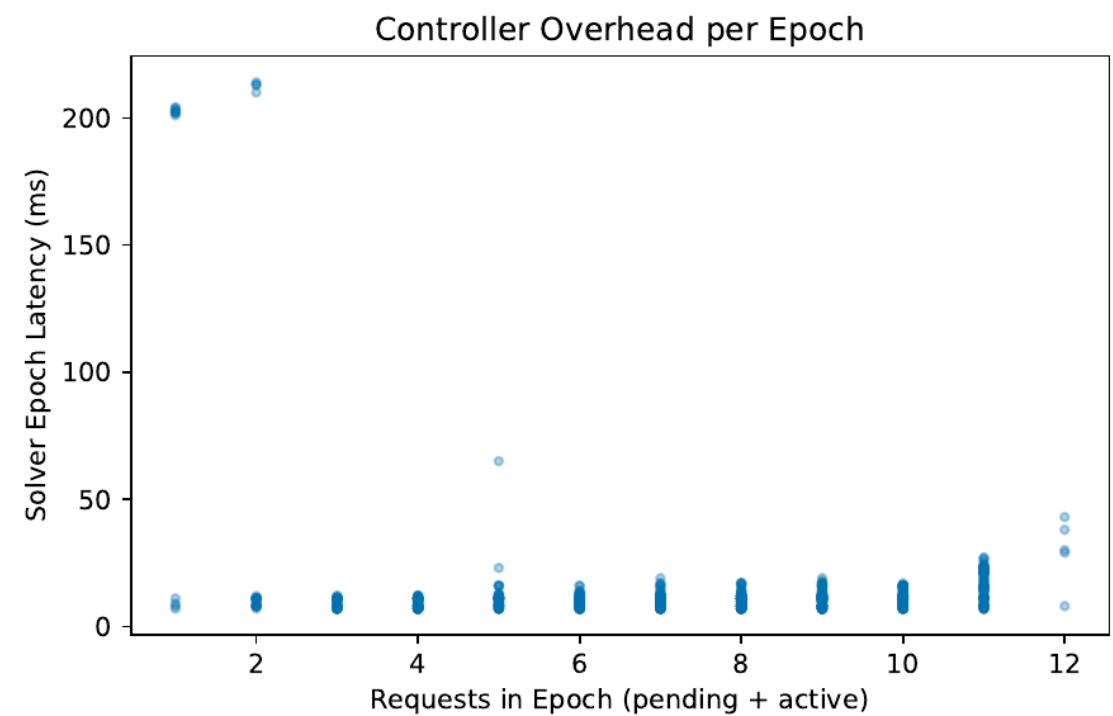
recovery_breakdown

Checkpoint Tradeoff



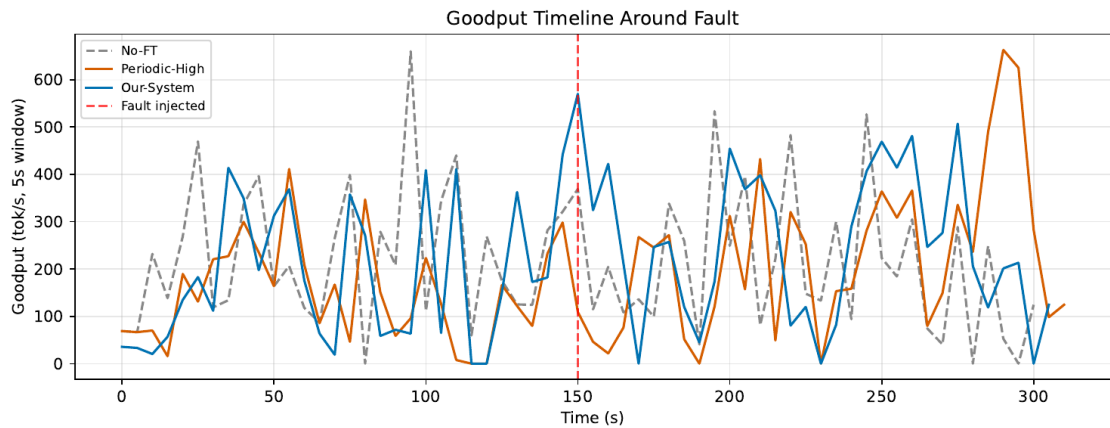
checkpoint_tradeoff

Controller Overhead



controller_overhead

Goodput Timeline (故障前后)



goodput_timeline

恢复率

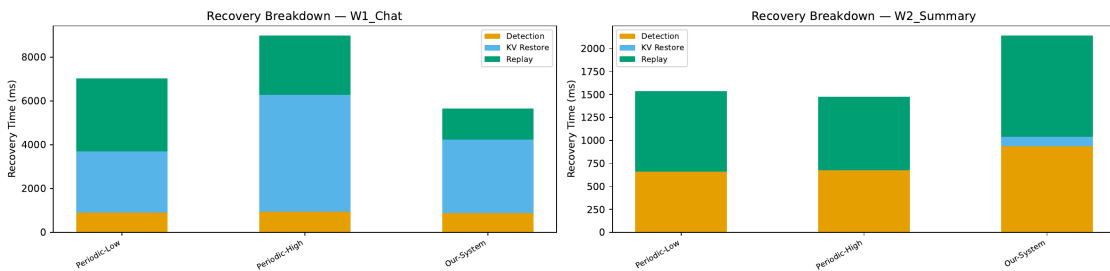
Baseline	恢复率范围	说明
No-FT	0-25%	基本不恢复
Periodic-Low	85-100%	大部分恢复
Periodic-High	47-100%	Heavy 负载下恢复率低
Our-System	100%	全部恢复

要点

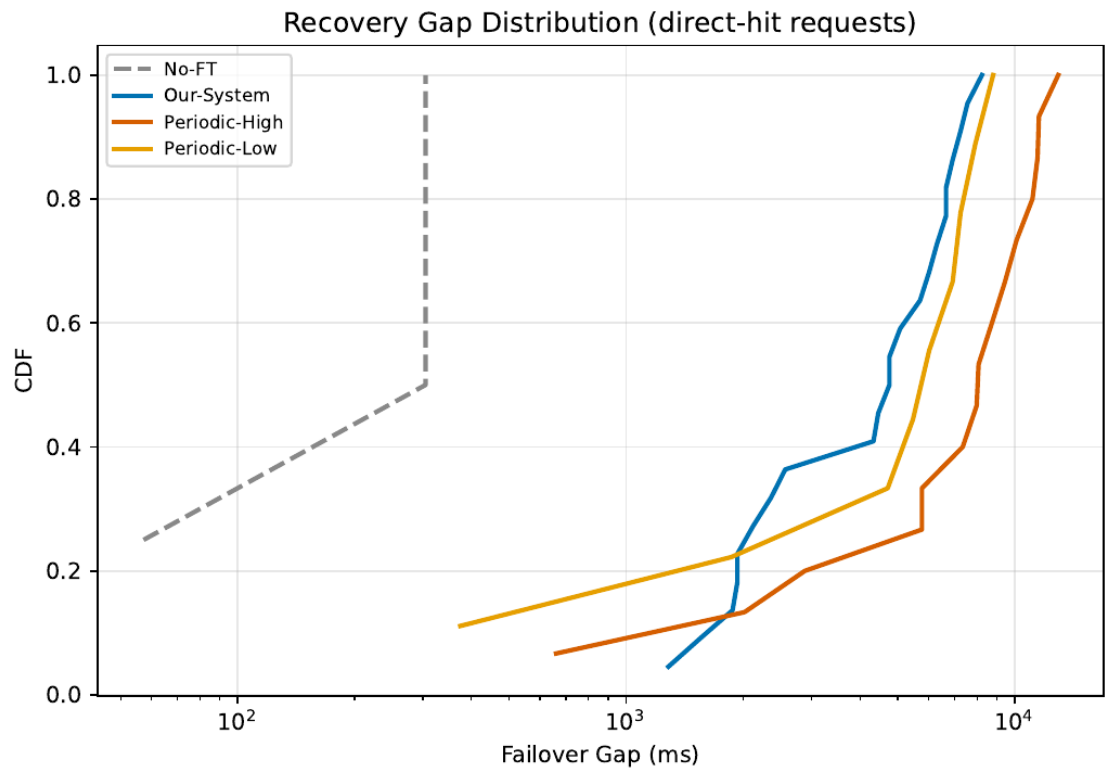
- 无故障：No-FT > Periodic-Low > Our-System $\approx$ Periodic-High。FT 开销 10-25%
- 有故障：
 - No-FT goodput 几乎不降，但恢复率 0%
 - **Periodic-High 故障后最差**（Heavy 只有 32 tok/s），checkpoint 太频繁拖慢恢复
 - Our-System 故障后稳定（109-111 tok/s），**恢复率 100%**
- **Our-System 的优势在 Heavy + 故障场景**：goodput 109.5 vs Periodic-High 32.0

3. E2_Recovery — 恢复时间分解 (8 runs)

Recovery Breakdown



Recovery Gap CDF



recovery_gap_cdf

数据

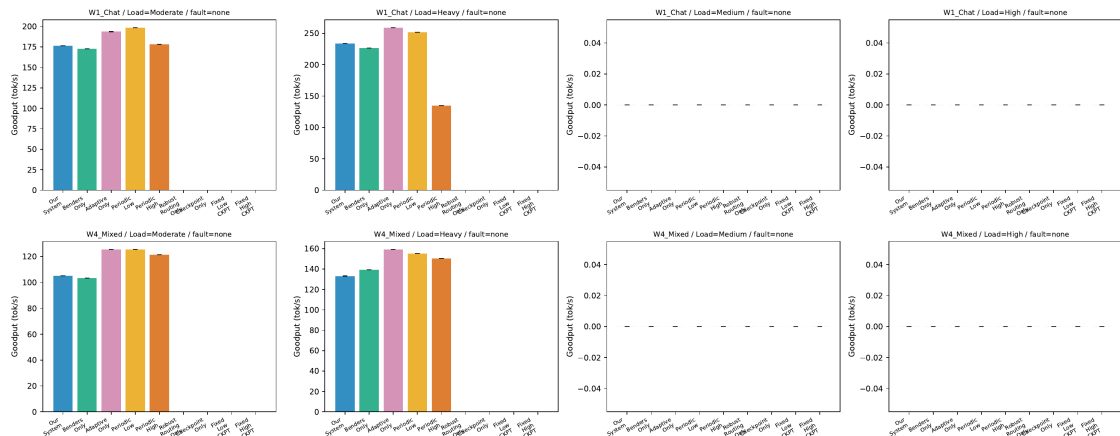
Baseline	Workload	Goodput	完成率	SLO Viol	TPOT P95	Gap P95	恢复率
No-FT	W1_Chat	219.8	98.0%	0.0%	37.7ms	—	0%
No-FT	W2_Summary	55.0	99.6%	0.0%	31.1ms	—	0%
Periodic-Low	W1_Chat	91.8	100.0%	53.2%	155.4ms	8495ms	100%
Periodic-Low	W2_Summary	53.9	100.0%	2.4%	84.3ms	374ms	100%
Periodic-High	W1_Chat	54.5	94.4%	70.1%	264.5ms	12177ms	85.7%
Periodic-High	W2_Summary	53.4	100.0%	3.2%	97.8ms	661ms	100%
Our-System	W1_Chat	106.9	99.6%	48.6%	129.7ms	7666ms	100%
Our-System	W2_Summary	46.9	99.2%	12.2%	100.6ms	2297ms	100%

要点

- **Periodic-High 反而比 Periodic-Low 差**：gap 12177ms vs 8495ms，恢复率 85.7% vs 100%。checkpoint 太频繁的开销 > 恢复收益
- **Our-System gap 最小**（W1_Chat 7666ms），但仍远超 3s gap SLO
- **W2_Summary 恢复快**（gap 374-2297ms）——短 decode 请求恢复容易

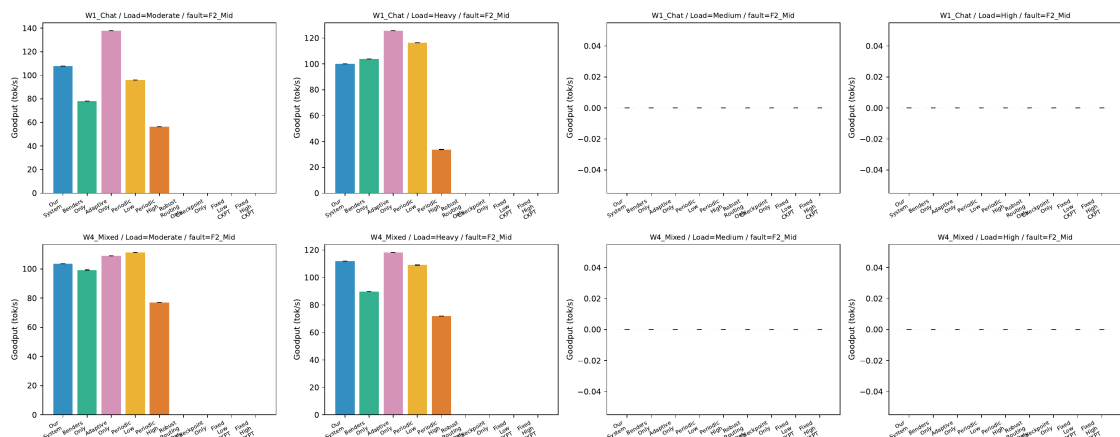
4. E3_Ablation — 消融实验 (40 runs)

Ablation 对比（无故障）



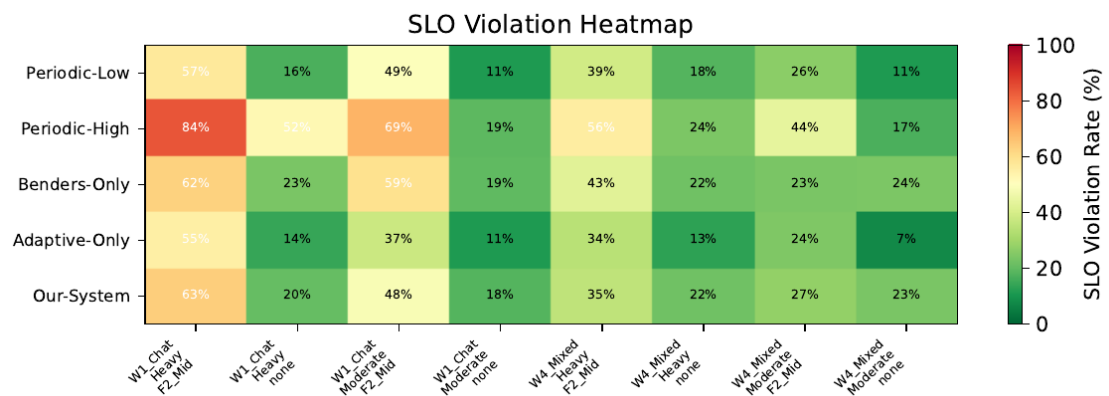
ablation_none

Ablation 对比（有故障 F2_Mid）



ablation_f2mid

Ablation Heatmap



ablation_heatmap

无故障 Goodput

Baseline	W1_Chat Mod	W1_Chat Heavy	W4_Mixed Mod	W4_Mixed Heavy
Periodic-Low	198.3	251.5	125.4	155.2
Periodic-High	178.3	134.8	121.5	150.4
Adaptive-Only	193.6	258.8	125.5	159.2
Benders-Only	172.6	226.5	103.4	139.2
Our-System	176.4	233.9	105.2	133.2

有故障 Goodput

Baseline	W1_Chat Mod	W1_Chat Heavy	W4_Mixed Mod	W4_Mixed Heavy
Periodic-Low	96.0	116.4	111.3	109.0
Periodic-High	56.3	33.8	76.9	71.8
Adaptive-Only	137.7	125.6	109.0	118.2
Benders-Only	78.2	103.9	99.3	89.7
Our-System	107.7	100.2	103.5	111.8

要点

- **Adaptive-Only 最强**：无故障 goodput 最高，有故障 W1_Chat 也最高（125-137）
- **Benders-Only 不如 Adaptive-Only**：solver 链路开销 > routing 收益（dp=2 下）
- **Our-System 介于两者之间**：Benders 开销部分抵消了自适应 ckpt 的收益
- **结论**：当前 dp=2 下，自适应 checkpoint 是主要贡献者，Benders routing 在更多 GPU 下才有价值

5. 总结

系统验证

- Checkpoint 正常 publish
- 故障恢复正常工作（Our-System 恢复率 100%）
- Solver 正常收敛

核心发现

1. **Our-System 恢复率 100%**，其他 baseline 最高 85-100%
2. **Periodic-High（每 1 block checkpoint）是最差的 FT 策略**——开销大、恢复慢
3. **自适应 checkpoint 是主要贡献者**（Adaptive-Only > Benders-Only）
4. **Benders solver 在 dp=2 下开销 > 收益**——需要 dp=4+ 展示 routing 价值

待解决

问题	现状	方向
SLO violation 高	所有 FT baseline TPOT 超 50ms	异步 checkpoint + 异步 solver
Failover gap 大	W1_Chat gap 7-12s, 超 3s SLO	更快的 restore 或更少的 replay
Solver 开销	dp=2 下得不偿失	扩展到 dp=4, 或异步化